MILITARY SPECIFICATION

CONNECTORS, TELECOMMUNICATION, POLARIZED SHELL, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for standard ribbon connectors for use in telecommunication applications.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-1183	-	Plastic Molding Material Acrylonitrile-Butadiene-Styrene (ABS), Rigid.
QQ-N-290	-	Nickel Plating (Electrodeposited).

MILITARY

MIL-M-14	-	Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-C-14550	-	Copper Plating (Electrodeposited).
MIL-G-45204	-	Gold Plating, Electrodeposited.
MIL-C-55330	-	Connectors, Preparation for Delivery of.
MIL-P-81728	-	Plating, Tin-Lead (Electrodeposited).

(See supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1344	-	Test Methods for Electrical Connectors.
MIL-STD-45662	-	Calibration Systems Requirements.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Electronic Support Division (AFLC), Gentile AF Station, Dayton OH 45444 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

American Society For Testing And Materials (ASTM)

ASTM-A-109 - Steel, Carbon, Cold-Rolled Strip, Specification for.

ASTM-B-103 - Phosphor Bronze Plate, Sheet, Strip, and Rolled Bar, Specification for.

ASTM-B-465 - Copper-Iron Alloy Plate, Sheet, Strip and Rolled Bar, Standard Specification for.

ASTM-B-633 - Zinc on Iron and Steel, Electrodeposited Coatings of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets (see 4.5.1). The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.3).

3.3 Materials (see 4.6.1). The materials shall be as specified herein (see 3.1). When a definite material is not specified, a material shall be used which will enable the connector to meet the performance requirements of this specification. Materials used in the construction of these connectors shall be fungus inert (see requirement 4 of MIL-STD-454). Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Body material. Unless otherwise specified (see 3.1), body material shall be a diallyl phthalate conforming to type MDG of MIL-M-14 or glass filled nonflammable material.

3.3.2 Contact material. Unless otherwise specified (see 3.1), contact material shall be phosphor bronze per ASTM-B-103, or copper iron (C19400) in accordance with ASTM-B-465.

3.3.3 Shell material. When applicable and unless otherwise specified (see 3.1), shell material shall be cold-rolled steel in accordance with ASTM-A-109, or plastic in accordance with MDG of MIL-M-14 or L-P-1183.

3.3.4 Dissimilar metals. Where dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. Dissimilar metals shall be as defined in MIL-STD-454, requirement 16.

3.3.5 Plating.

3.3.5.1 Overall finish. Contacts shall be overall gold-plated, 30 microinches minimum thickness in accordance with MIL-G-45204, type I, grade C, class 1 over a nickel underplate of 0.000030 to 0.000150 inch thick in accordance with QQ-N-290. Silver shall not be used as a underplate.

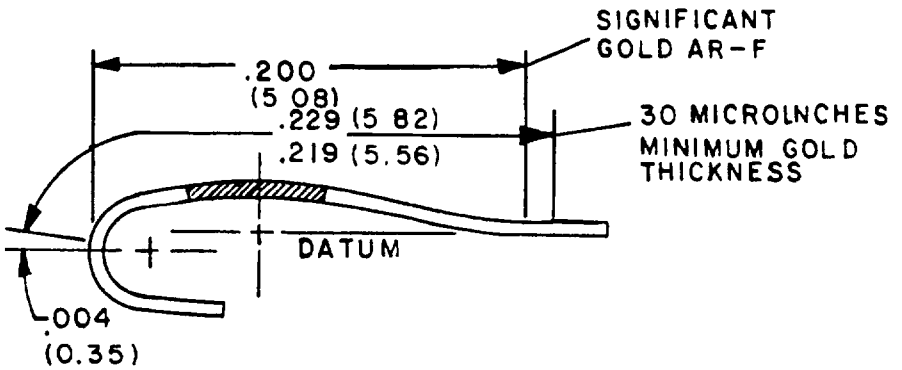
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3.3.5.1.1 Localized finished. Contacts shall have a nickel underplate in accordance with QQ-N-290, class 2.

3.3.5.1.2 Mating area. The contact mating area (see figure 1) shall be gold-plated 30 microinches minimum thickness in accordance with MIL-G-45204, type I, grade C, class 1 over copper flash in accordance with MIL-C-14550 or nickel underplate in accordance with QQ-N-290 (see 3.1).

3.3.5.1.3 Terminations or wire connection end. Terminations shall be plated with 100 microinches tin-lead in accordance with MIL-P-81728, 50 to 95 percent tin (see 3.1).

3.3.5.2 Shell plating. When applicable and unless otherwise specified (see 3.1), connector shells shall be zinc plated in accordance with ASTM-B-633, type RS, followed by a clear chromate conversion in accordance with ASTM-B-633 class 3, type II. Plating shall be .0002 to .0004 inch.



NOTE:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Contact finish.

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3.3.6 Restricted materials.

3.3.6.1 Flammable, explosive, or toxic. Material shall be nonflammable, nonexplosive, and nontoxic over the operating temperature range.

3.3.6.2 Corrosion resistance. Connectors and accessories shall be corrosion resistant materials or treated to prevent corrosion.

3.3.6.3 Ferrous. Material containing more than 5 percent iron shall not be used for current carrying parts.

3.4 Design and construction (see 4.6.1). Connectors shall be of the design, construction, and physical dimensions specified herein (see 3.1).

3.4.1 Threaded parts. Unless otherwise specified (see 3.1), all threaded parts shall be in accordance with FED-STD-H28. Thread forming screws may be used where specified (see 3.1).

3.4.1.1 Engagement of threaded parts. All threaded parts shall engage by at least two full threads.

3.4.2 Connector assembly. No parts of the connector assembly shall be permanently displaced from their original, normal, or fitted position (see 3.1) at completion of the specified tests.

3.4.3 Contact cavities. The arrangement of contact cavities in the insulator body of connectors shall be as specified (see 3.1). Each contact cavity shall be fitted with a contact assembly so confined in the cavity that accidental removal is prevented, and positive alignment of the respective contacts, including the contact termination, is effected.

3.4.3.1 Contact clearance. The tops of the sockets in the uppermost position shall be below the insert face.

3.4.4 Contact identification. Contact positions shall be identified by legible letters or numbers, molded or stamped on the connector body as specified (see 3.1).

3.4.5 Body design. Connector bodies shall be designed and constructed with proper sections and radii so that they will not crack, chip, or break in assembly or normal service. The insulator of each plug and receptacle shall be of one piece construction. Depressions, when used to achieve longer creepage paths, shall not cause structural weakness.

3.4.6 Polarization. A polarization feature to assure correct insertion shall be incorporated in each connector assembly as specified (see 3.1).

3.4.7 Alignment. Each connector shall have a feature which will insure proper alignment of contact before mating.

3.4.8 Interchangeability. Receptacles of a given type shall be capable of being mated with associated plugs meeting the requirements of this specification. The mated connectors and individual plugs and receptacles having related part numbers shall be directly and completely interchangeable with each other with respect to installation and performance as specified herein.

3.5 Performance.

3.5.1 Solderability. Unless otherwise specified (see 3.1), contacts shall withstand the test specified in 4.6.2.

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3.5.2 Mating and unmating forces. When tested as specified in 4.6.3, the maximum and minimum mating and unmating forces shall be as specified in table I.

TABLE I. Mating and unmating forces.

Number of contacts	Mating (maximum)	Unmating (minimum)
14	7 pounds	2 pounds
24	12 pounds	4 pounds
36	18 pounds	6 pounds
50	25 pounds	7 pounds
64	32 pounds	8 pounds

3.5.3 Continuity. When tested in accordance with 4.6.4, all contacts of the test connector shall electrically contact the continuity gauge.

3.5.4 Contact resistance. When tested in accordance with 4.6.5, the contact resistance shall not exceed 0.006 ohm. The current shall be specified by MIL-STD-1344, method 3002.1, low-signal level contact resistance.

3.5.5 Contact retention. When tested as specified in 4.6.6, the contacts shall withstand an axial load of 2 pounds minimum. The axial displacement of contacts shall not exceed 0.015 inch (0.38 mm) while under load.

3.5.6 Dielectric withstanding voltage. When tested as specified in 4.6.7, connectors shall withstand a minimum of 500 volts root mean square (rms) without dielectric breakdown or flashover.

3.5.7 Insulation resistance. When tested in accordance with 4.6.8, the initial insulation resistance shall not be less than 5,000 megohms.

3.5.8 Durability. When tested in accordance with 4.6.9, connectors shall show no evidence of cracking or breaking, the contact resistance requirements of 3.5.4 shall not be exceeded, and the mating and unmating requirements of 3.5.2 shall be met.

3.5.9 Salt spray (corrosion). When mated connector pairs are tested in accordance with 4.6.10, there shall be no peeling, chipping, or blistering of metal surfaces or exposure of base metal, and the connectors shall conform to the requirements of 3.5.4 and 3.5.7.

3.5.10 Temperature cycling. When a connector is tested in accordance with 4.6.11, there shall be no evidence of cracking or crazing of the connector body or other physical damage to the connector assembly. Change in contact resistance (see 3.5.4) shall not be more than 25 percent of the value specified.

3.5.11 Humidity. When tested in accordance with 4.6.12, insulation resistance shall be greater than 1,000 megohms.

3.5.12 Crush (when specified, see 3.1). When tested in accordance with 4.6.13, the mated connectors (cable to cable type) shall be capable of withstanding a force of 1,500 pounds. Following crush test, the connectors shall conform to the requirements of 3.5.6 and 3.5.11.

3.6 Marking (see 4.6.1). Connectors shall be marked in accordance with method I of MIL-STD-1285, and shall include the manufacturer's name, trademark, source code, or Manufacturer's Federal Supply Code (FSCM) and the military part number (see 3.1).

3.7 Workmanship (see 4.6.1). Connectors shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, and appearance. There shall be no evidence of loose contacts, poor or improper molding or fabrication, damaged or improperly assembled contacts, peeling, flaking, or chipping of plating or finish, mechanical damage due to testing environment; nicks or burrs of metal parts or surfaces, improper or incorrect marking, or improper tinning of solder cups, terminals, pins, or contacts.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-1344.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. Six mated pairs of connectors with the largest number of contacts, and two mated pairs of each of the other part numbers from each specification sheet shall be subjected to first article inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided equally into two groups of four units each, and subjected to the inspections for their particular group.

4.4.3 Failures. Any failures shall be cause for refusal to grant lot approval.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Inspection lot. An inspection lot shall consist of all connectors of the same basic type product under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of visual and mechanical inspection (see 4.6.1).

4.5.1.2.1 Sampling plan (group A inspection). Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be 1.0 and 4.0 (percent defective) for major or minor defects, respectively. Major and minor defects shall be as defined in MIL-STD-105.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I (all sample units)		
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.6.1
Solderability	3.5.1	4.6.2
Mating and unmating forces	3.5.2	4.6.3
Continuity	3.5.3	4.6.4
Contact resistance	3.5.4	4.6.5
Contact retention	3.5.5	4.6.6
Dielectric withstanding voltage	3.5.6	4.6.7
Insulation resistance	3.5.7	4.6.8
Group II (1/2 sample units).		
Durability	3.5.8	4.6.9
Contact resistance	3.5.4	4.6.5
Mating and unmating forces	3.5.2	4.6.3
Salt spray (corrosion)	3.5.9	4.6.10
Contact resistance	3.5.4	4.6.5
Insulation resistance	3.5.7	4.6.8
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.6.1
Group III (1/2 sample units)		
Temperature cycling	3.5.10	4.6.11
Contact resistance	3.5.4	4.6.5
Humidity	3.5.11	4.6.12
Insulation resistance	3.5.7	4.6.8
Crush (when specified)	3.5.12	4.6.13
Dielectric withstanding voltage	3.5.6	4.6.7
Continuity	3.5.3	4.6.4
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.6.1

4.5.1.2.2 Rejected lots (group A inspection). If an inspection is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirements of correction is disclosed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III, in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Method paragraph
Mating and unmating forces	3.5.2	4.6.3
Continuity	3.5.3	4.6.4
Contact retention	3.5.5	4.6.6
Dielectric withstanding voltage	3.5.6	4.6.7

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4.5.1.3.1 Sampling plan (group B inspection). The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-1. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL for mating and unmating forces, and contact retention shall be 1.0 percent defective. The AQL for continuity shall be 2.5 percent defective. The AQL for dielectric withstanding voltage shall be 4 percent defective.

4.5.1.3.2 Rejected lots (group B inspection). If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all group B inspection may be delivered on the contract if the lot is accepted and the sample units are still within specified electrical tolerances.

4.5.2 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of MIL-C-55330.

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection (see 3.1, 3.3, 3.4, 3.6, and 3.7). Connectors shall be inspected to verify that the materials, screw threads, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.2 Solderability (see 3.5.1). Each contact will be subjected to method 206 of MIL-STD-202.

4.6.3 Mating and unmating forces (see 3.5.2). Mating and unmating forces shall be measured in accordance with method 2013 of MIL-STD-1344. Mating and unmating forces shall conform to the requirements of 3.5.2.

4.6.4 Continuity (see 3.5.3). After three full insertions of the appropriate sizing gauge (see table IV and figures 2 and 4) all contacts of the connector plug or receptacle shall make electrical contact with a fully inserted continuity gauge (see table IV and figures 3 and 5). If the specified gauge fails to make electrical contact on the first insertion, the gauge may be reinserted a second or third time; however, all contacts shall make electrical contact on the third insertion. Electrical continuity shall be determined by an electrical continuity test that applies an open circuit voltage of not more than 10 volts. Connectors shall conform to the requirements of 3.5.3.

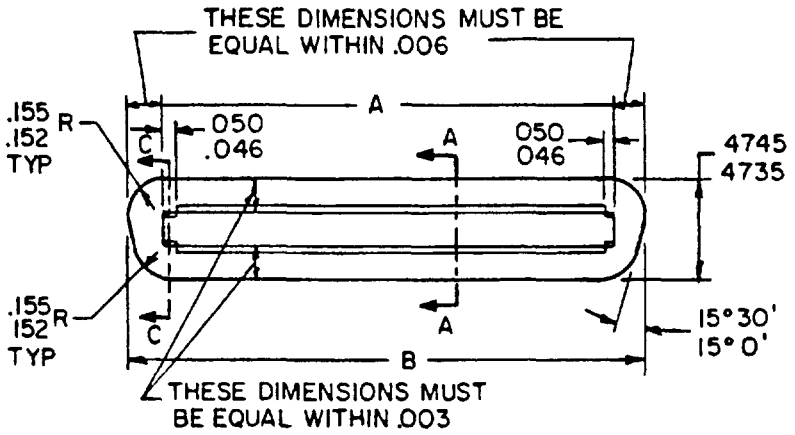
4.6.5 Contact resistance (see 3.5.4). The contact resistance shall be measured individually between each mated pair of contacts. A minimum of seven mated pairs of contacts shall be measured on each test specimen in accordance with method 3004 of MIL-STD-1344 with the following details and exceptions:

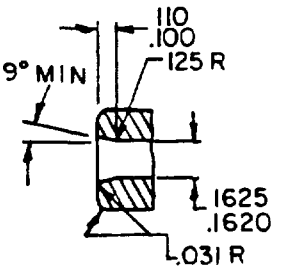
a. Attach current-voltage leads at extreme ends of contacts.

b. Test current: See rated current (3.1).

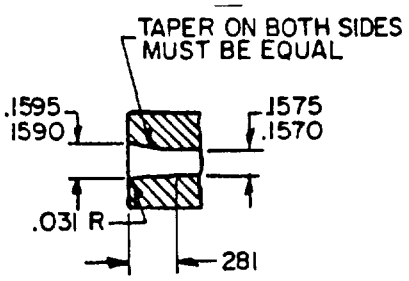
c. Measurements shall conform to the requirements of 3.5.4.

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SECT A-A



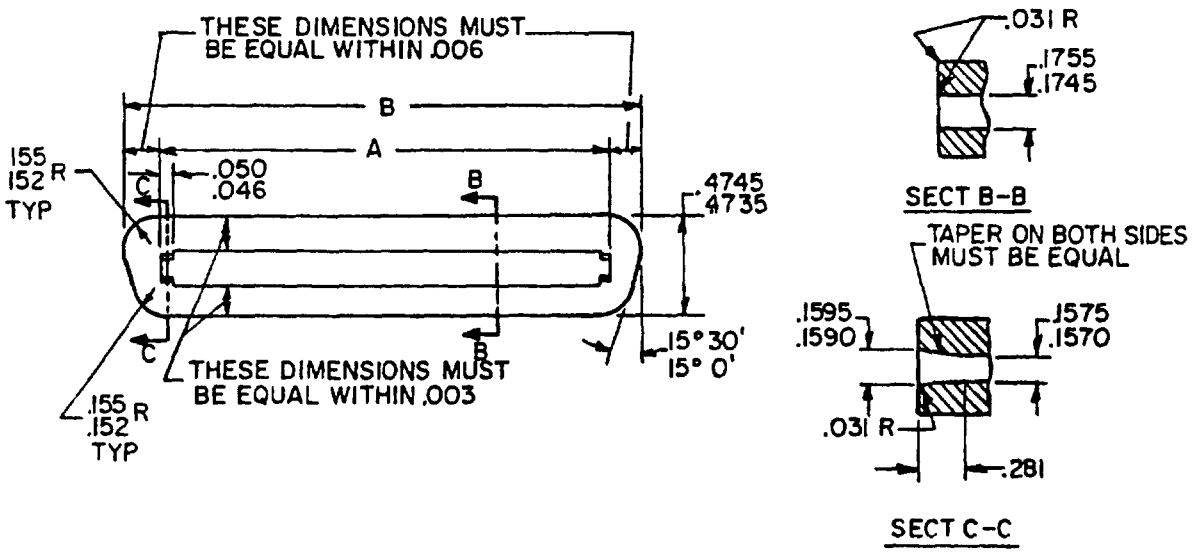
SECT C-C

INCHES	MM	INCHES	MM
.003	0.08	.1570	3.988
.006	0.15	.1575	4.000
.031	0.79	.1590	4.039
.046	1.17	.1595	4.051
.050	1.27	.1620	4.115
.100	2.54	.1625	4.128
.110	2.79	.281	7.14
.125	3.18	.4735	12.027
.152	3.86	.4745	12.052
.155	3.94		

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified tolerance shall be ±0.001 inches

FIGURE 2. Plug sizing gauge

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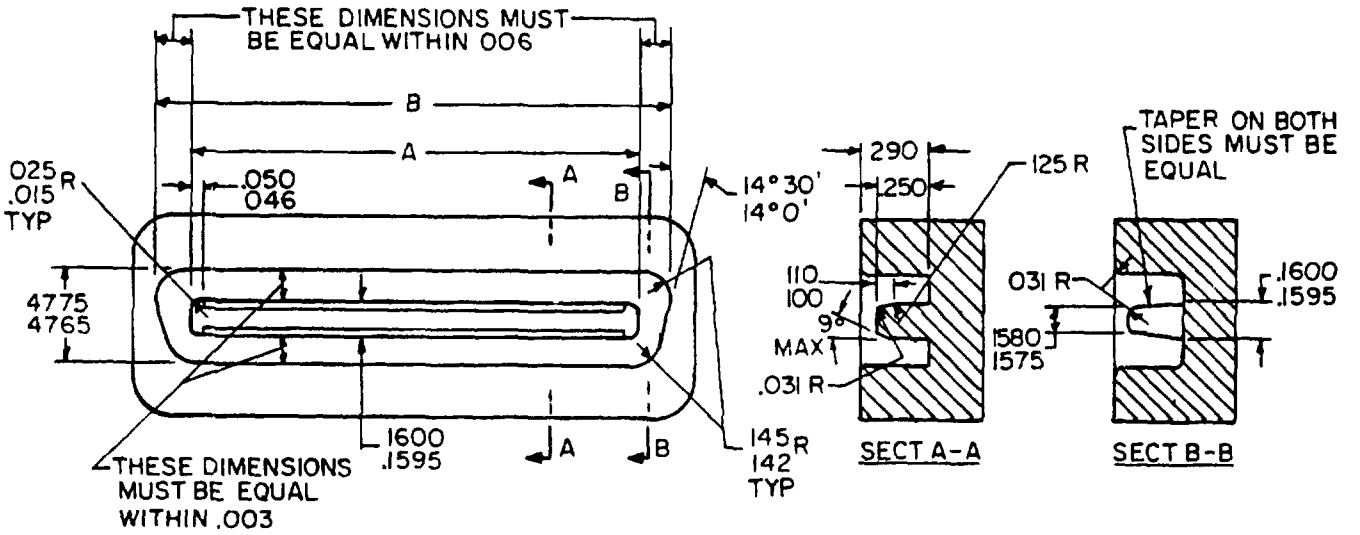


INCHES	MM	INCHES	MM
.003	0.08	.1575	4.000
.006	0.15	.1590	4.039
.031	0.79	.1595	4.051
.046	1.17	.1745	4.432
.050	1.27	.1755	4.458
.152	3.86	.281	7.14
.155	3.94	.4735	12.027
.1570	3.988	.4745	12.052

NOTES.
1 Dimensions are in inches.
2 Metric equivalents are given for general information only
3 Unless otherwise specified tolerance shall be ±0 001 inches

FIGURE 3. Plug continuity gauge.

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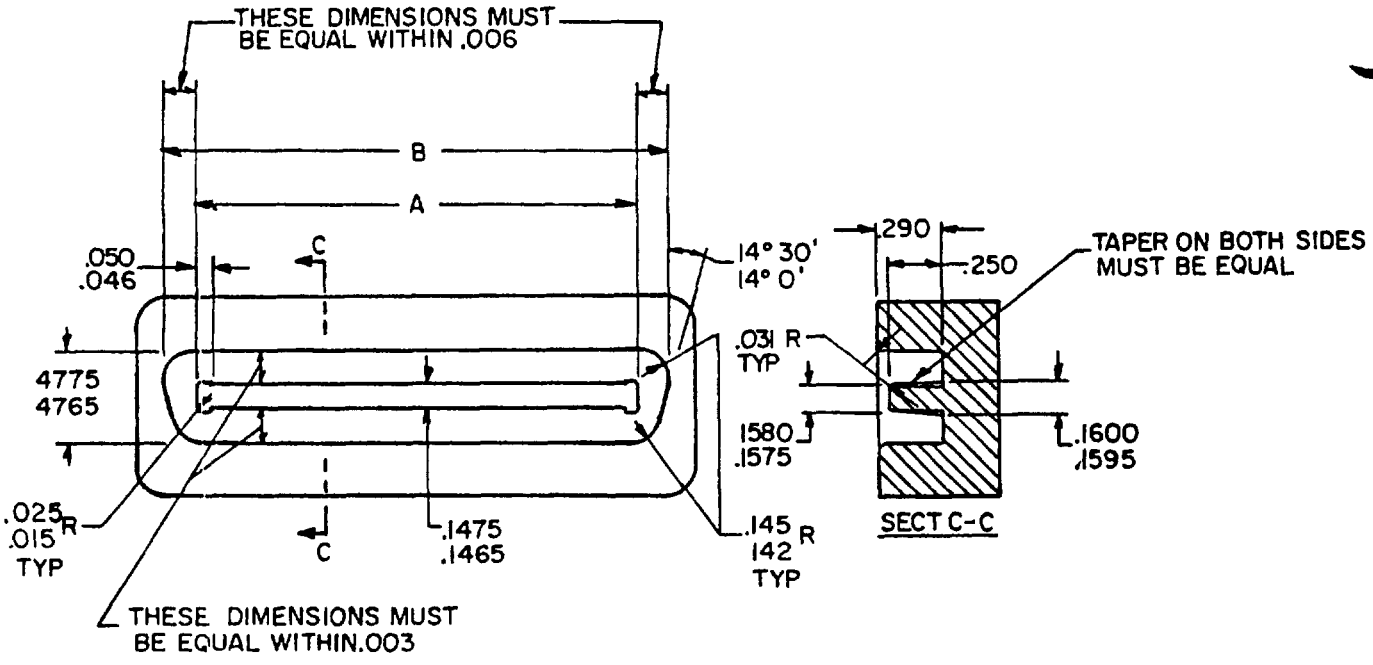


INCHES	MM	INCHES	MM
.003	0.08	.145	3.68
.006	0.15	.1575	4.000
.015	0.38	.1580	4.013
.025	0.64	.1595	4.05
.031	0.79	.1600	4.064
.046	1.17	.250	6.35
.050	1.27	.290	7.37
.125	3.18	4765	12.103
.142	3.61	.4775	12.128

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified tolerance shall be ±0 001 inches

FIGURE 4. Receptacle sizing gauge.

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INCHES	MM	INCHES	MM
.003	0.08	.1465	3.721
.006	0.15	.1475	3.746
.015	0.38	.1575	4.000
.025	0.64	.1580	4.013
.031	0.79	.1595	4.05
.046	1.17	.1600	4.064
.050	1.27	.250	6.35
.142	3.61	.290	7.37
.145	3.68	.4765	12.103
		.4775	12.128

NOTES.
1 Dimensions are in inches.
2 Metric equivalents are given for general information only.
3 Unless otherwise specified tolerance shall be ±0.001 inches.

FIGURE 5. Receptacle continuity gauge.

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4.6.6 Contact retention (see 3.5.5). Connectors shall be tested in accordance with method 2007 of MIL-STD-1344 with the following details and exceptions

a. Number of samples Minimum of seven contacts per test specimen.

b. Applied axial load 1 pound per second until a 2 pound load has been reached. Maintain load for 5 seconds. (When hookup wire is used, the wire shall be large enough to withstand the applied load).

c. Axial direction Straight.

d. Measurements shall conform to the requirements of 3.5.5.

4 6.7 Dielectric withstanding voltage (see 3.5.6) Connectors shall be tested in accordance with method 3001 of MIL-STD-1344. The following details shall apply·

a. Number of samples· Minimum of seven contacts per test specimen.

b. Magnitude of test voltage and nature of potential 500 volts ac.

c. Duration of application· For quality conformance inspection, the test voltage shall be applied at a rate of 100 volts per second and the voltage may be applied instantaneously and shall be maintained for at least 5 seconds.

d. Points of application· Between mutally insulated contacts of the connector.

4.6.8 Insulation resistance (see 3.5.7). Unmated connectors shall be tested for insulation resistance in accordance with method 3003 of MIL-STD-1344, with the following details and exceptions.

a. Number of samples Minimum of seven contacts per test specimen.

b. Measurements shall be made between adjacent pairs of contacts.

c. A test potential of 500 volts dc ±10 percent shall be applied for a period of 1 minute.

d. Measurements shall conform to the requirements of 3.5.7.

4.6.9 Durability (see 3.5.8). Connectors shall be tested in accordance with method 2016, of MIL-STD-1344, with the following details and exceptions

a. The connectors, plug and receptacle, shall be mated and unmated 200 times in a manner that will simulate actual service.

b. After completion of test, the connectors shall conform to the requirements of 3.5.8.

4.6.10 Salt spray (see 3.5.9). Connectors shall be tested in accordance with method 1001 of MIL-STD-1344, with the following details and exceptions·

a. Applicable salt solution. 5 percent.

b. Test condition B.

c. Upon completion of salt spray testing, connectors shall conform to the requirements of 3.5.9.

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TABLE IV. Sizing and continuity gauges.

Unit	Sizing			Continuity			
	Dim A		Dim B	Dim A		Dim B	
	Min	Max	Ref	Min	Max	Min	Max
Plug							
14 Contact	.6835 (17.361)	.6855 (17.412)	.855 (21.72)	.6853 (17.361)	.6855 (17.412)	.9985 (25.362)	.9995 (25.387)
24 Contact	1.1085 (28.156)	1.1105 (28.207)	1.280 (32.51)	1.1085 (28.156)	1.1105 (28.207)	1.4235 (36.157)	1.4245 (36.182)
36 Contact	1.6185 (41.110)	1.6205 (41.161)	1.790 (45.47)	1.6185 (41.110)	1.6205 (41.161)	1.9295 (49.009)	1.9305 (49.035)
50 Contact	2.2135 (56.223)	2.2155 (56.274)	2.385 (60.58)	2.2135 (56.223)	2.2155 (56.274)	2.5285 (64.224)	2.5295 (64.249)
Receptacle							
14 Contact	.6925 (17.590)	.6945 (17.640)	.858 (21.79)	.6925 (17.590)	.6945 (17.640)	1.0015 (25.438)	1.0025 (25.464)
24 Contact	1.1175 (28.384)	1.1195 (28.435)	1.286 (32.66)	1.1175 (28.384)	1.1195 (28.435)	1.4295 (36.309)	1.4305 (36.335)
36 Contact	1.6275 (41.339)	1.6295 (41.389)	1.793 (45.54)	1.6275 (41.339)	1.6295 (41.389)	1.9365 (48.187)	1.9375 (49.212)
50 Contact	2.2225 (56.452)	2.2245 (56.502)	2.388 (60.65)	2.2225 (56.452)	2.2245 (56.502)	2.5315 (64.300)	2.5325 (64.326)

Metric equivalents are in parentheses.

4.6.11 Temperature cycling (see 3.5.10). Connectors shall be tested in accordance with method 1003.1 of MIL-STD-1344, with the following details and exception.

After temperature cycling conditioning, the connectors shall conform to the requirements of 3.5.10.

4.6.12 Humidity (see 3.5.11). Connectors shall be tested in accordance with method 1002, type II of MIL-STD-1344, with the following details and exceptions:

a. Connectors shall be wired as specified in 4.6.7 (dielectric withstanding voltage).

b. Duration of test: 96 hours, steps 7a and 7b shall not be required.

c. Connectors shall conform to the requirements of 3.5.11 upon completion of humidity conditioning and removal of surface moisture from connectors.

4.6.13 Crush (see 3.5.12). Mated plug and receptacles shall be tested in accordance with method 2008 of MIL-STD-1344. Following the test, connectors shall conform to the requirements of 3.5.12.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330.

6. NOTES

6.1 Intended use. These connectors are designed for use in telecommunication equipment, particularly as an interface between computerized equipment and telephone lines. These connectors may also be used in such systems as satellite ground stations, aircraft ground test equipment, and simulators. These connectors are not recommended for use in equipment that experiences medium to high vibration or shock.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b Title, number, and date of the applicable specification sheet, and the complete part number (see 3.1).

6.3 First article. When a first article inspection is required, the item shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be preproduction sample, a first production item, a sample selected from the first lot of production items, or it may be a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one, two, etc. unit(s). The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examinations, test approval of the documents' first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Air Force - 99
DLA - ES

User activities:
Air Force - 11, 17, 19

Agent
DLA - ES

Preparing activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-83515A	Connectors, Telecommunication, Polarized Shell, Gen. Spec. For

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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